

Fiat / Abarth 124 Spider 2017+ Apple CarPlay / Android Auto Upgrade. Part 2 – Hardware Install.



This is part two of a guide covering the steps required to install the 'official' Mazda upgrade to the CMU (what we used to call 'the stereo') to add Apple CarPlay and Android Auto functionality in your 2017+ FIAT or Abarth 124 Spider. This part of the guide covers installing necessary replacement hardware in your dashboard. ***The first part of this guide covers updating the firmware in your CMU and must be carried out prior to the steps described in this part of the guide.***

WHAT YOU WILL NEED:

For this part of the install, you will need:

- Replacement USB cables: Mazda Part C922-V6-605A
- Replacement USB HUB: Mazda Part TK78-66-9U0C
- Metric socket set with 10mm socket and 200mm / 8" extension.
- Cable ties (approximately 10)
- Scissors
- Flat head and Philips screwdrivers.
- Foam Tape
- Trim removal tools

I ordered the required USB hub and cables on Amazon at

https://www.amazon.com/gp/product/B07KRPSRKH/ref=ppx_yo_dt_b_asin_title_o02_o00_s00?ie=UTF8&p_sc=1 – price at time of writing was \$163.

The foam tape is used secure the excess USB cable behind the dash and to prevent it rattling (the same replacement USB cables are used for other Mazda vehicles, all of which are significantly larger than the 124 / MX-5 and need a longer cable run so there's quite a bit of excess). I ordered the following:

https://www.amazon.com/gp/product/B01G382DX4/ref=ppx_yo_dt_b_asin_title_o01_o00_s00?ie=UTF8&p_sc=1 Others have used: <https://tinyurl.com/yatbqfn9> As it turns out, the hardware kit I ordered with the replacement USB hub and cables included foam tape and cable ties so this turned out to be unnecessary.

Trim removal tools simply make it less likely you'll damage the bits of trim that need to be removed to access the CMU. I have this kit:

https://www.amazon.com/gp/product/B01L8GHB7O/ref=ppx_yo_dt_b_asin_title_o04_o00_s00?ie=UTF8&p_sc=1

You can also use a flat headed screwdriver with some electrical tape over the head (to protect the trim) for anything we'd be using a trim tool for.

USEFUL LINKS AND ACKNOWLEDGEMENTS

The information in this how-to has been pulled from a number of sources and I hereby nod my hat to:

Ameridan and his fantastic blog and work on the All-In-Tweaks at <https://21stcenturyfiat124spider.wordpress.com> This is well worth checking out even if you never have any intention of doing the CarPlay / Android Auto update.

ASH8 for his original Mazda how-to thread at <https://mazda3revolution.com/forums/2014-2018-mazda-3-skyactiv-audio-electronics/230249-genuine-carplay-android-auto-all-here-now.html>

I was inspired by rsmagee and TheSalamander – Fiat 124 CarPlay pioneers – in this thread: <https://www.124spider.org/forum/26-electronics-audio-lighting/30666-carplay-up-running.html>

The original Mazda All-In-One tweaks are at <https://mazdatweaks.com/>

In order to support CarPlay and Android Auto, the USB hub (the little area of the dash where you plug in USB cables and your navigation SD card) needs to be replaced along with the cables connecting the CMU to the USB hub. The replacement hub can be identified by the phone icon next to one of the USB ports – you use this port to connect your phone if you want to run CarPlay or Android Auto. It's unclear to me exactly what the replacement hub adds, but I've heard people say it provides higher voltage USB connectivity (i.e. faster charging or stopping the battery draining while your phone is running CarPlay or Android Auto) and /or it correctly identifies the system as being licensed to run CarPlay or Android Auto.

The photos below show the replacement parts we will be fitting including the foam pads and cable ties my parts kit included.



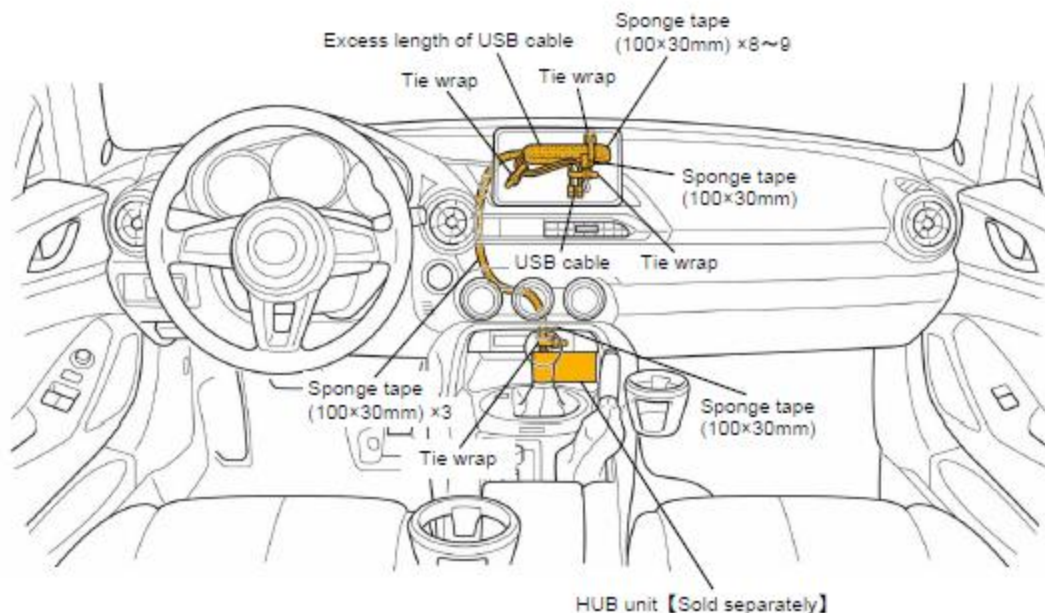


There are two USB cables in the kit, one with a gray and blue plug at one end and a brown plug at the other end, and one with a gray and green plug at one end and a black plug at the other end. Make sure you have both cables.

The foam padding comes already separated into 30mm by 100mm sections:



The following diagram shows the routing of the cables we will be installing:



In order to be able to access the back of the CMU and to replace the hub unit, we need to remove a number of pieces of trim. This will seem intimidating at first, but in reality, it's pretty straight forward. Most pieces simply pull off and clip back on.

A full list of shop manual instructions for each piece of trim to be removed, as well as the Mazda cable installation instructions and videos of an installation can be found at:

[https://my.hidrive.com/share/hsodpqja.l#\\$/Mazda_Firmware/-CarPlay-AndroidAuto%20INSTALL](https://my.hidrive.com/share/hsodpqja.l#$/Mazda_Firmware/-CarPlay-AndroidAuto%20INSTALL)

Eddie Clark also posted a video of his CarPlay installation on his YouTube Pistons and Petrol channel:

<https://youtu.be/Qcyan28QXs4>

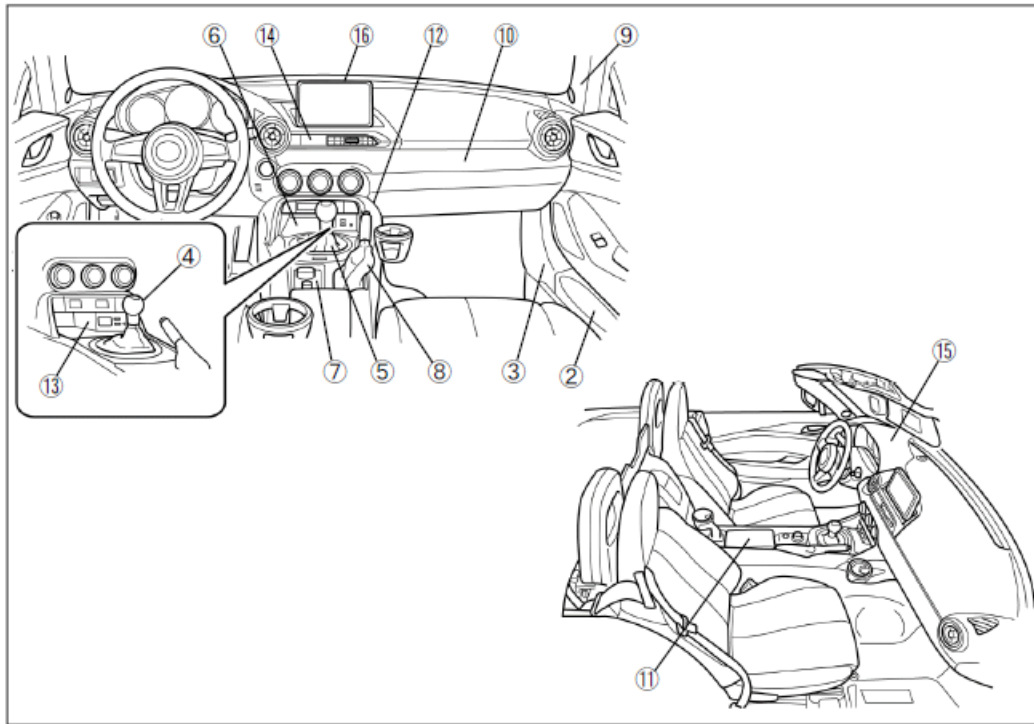
Trim pieces will be removed in the following order:

1. Passenger Side Scuff Plate
2. Passenger Side Front Side Trim
3. Shift knob
4. Shift Panel, Console Panel and Upper Panel (these are listed separately in the official Mazda instructions, but can actually be removed as one piece)
5. Parking brake boot panel
6. Passenger side A-pillar trim
7. Passenger side lower trim
8. Rear console
9. Front console and Front Console Panel – again, these are listed separately in the Mazda instructions but can be removed as one piece.
10. Center Panel No. 2

11. Meter (Gauge Cluster) Hood

12. Center display (this is the display screen and actual CMU unit).

The following diagram shows the location of each item:

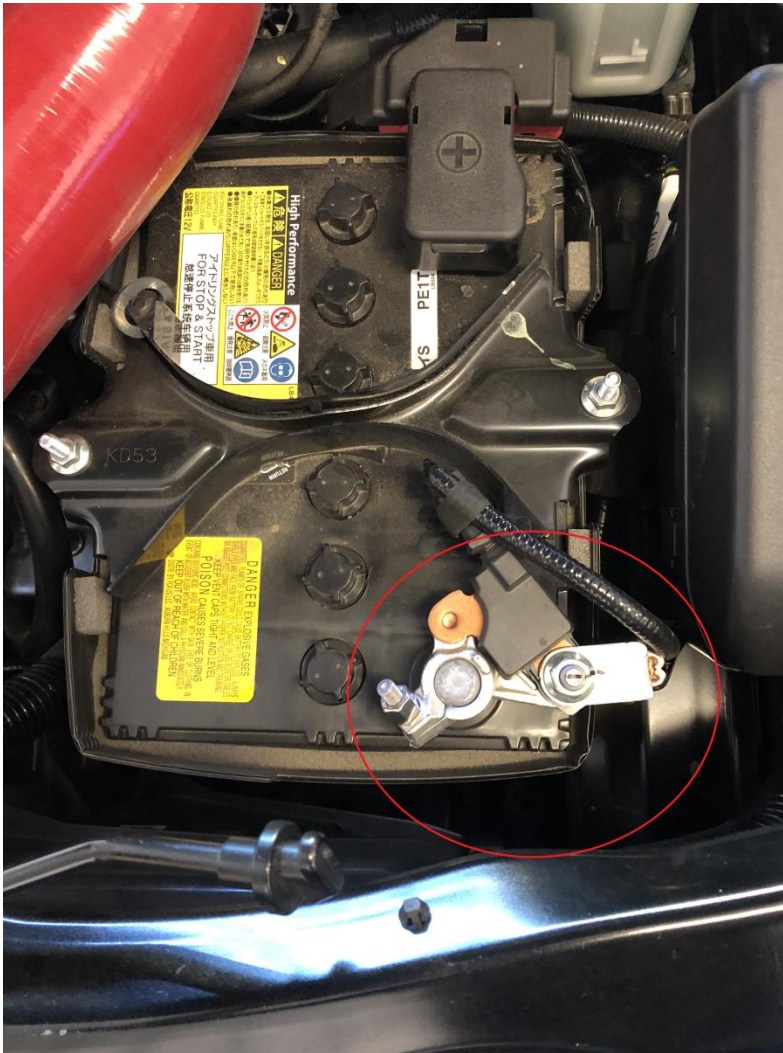


- ① Negative battery cable disconnection
- ② Passenger-side scuff plate
- ③ Passenger-side front side trim
- ④ Shift lever knob (MT) or Selector lever knob (AT)
- ⑤ Shift panel
- ⑥ Console panel
- ⑦ Upper panel
- ⑧ Parking brake lever boots panel
- ⑨ Passenger-side A-pillar trim
- ⑩ Passenger-side lower panel
- ⑪ Rear console
- ⑫ Front console panel
- ⑬ Front console
- ⑭ Center panel No.2
- ⑮ Meter hood
- ⑯ Center display

Overall, it took me roughly two and half hours to complete the installation, including stopping to take pictures.

STEP 1 – TRIM REMOVAL

1. Start by lowering the convertible top and windows. While this isn't entirely necessary, I find it much easier to work with the interior of the car as exposed as possible.
2. Make sure your car is parked somewhere where you can fully open both doors.
3. Designate a space for your removed trim pieces. I laid out a packing blanket on my garage floor to put the pieces on. You may also want to put a towel or similar down on the floor beside each door to kneel on as you will be working in those areas.
4. Use a 10mm socket to disconnect the negative terminal of your battery. Note, the black plastic piece on the terminal is rather fragile – do not strike or pull on it:



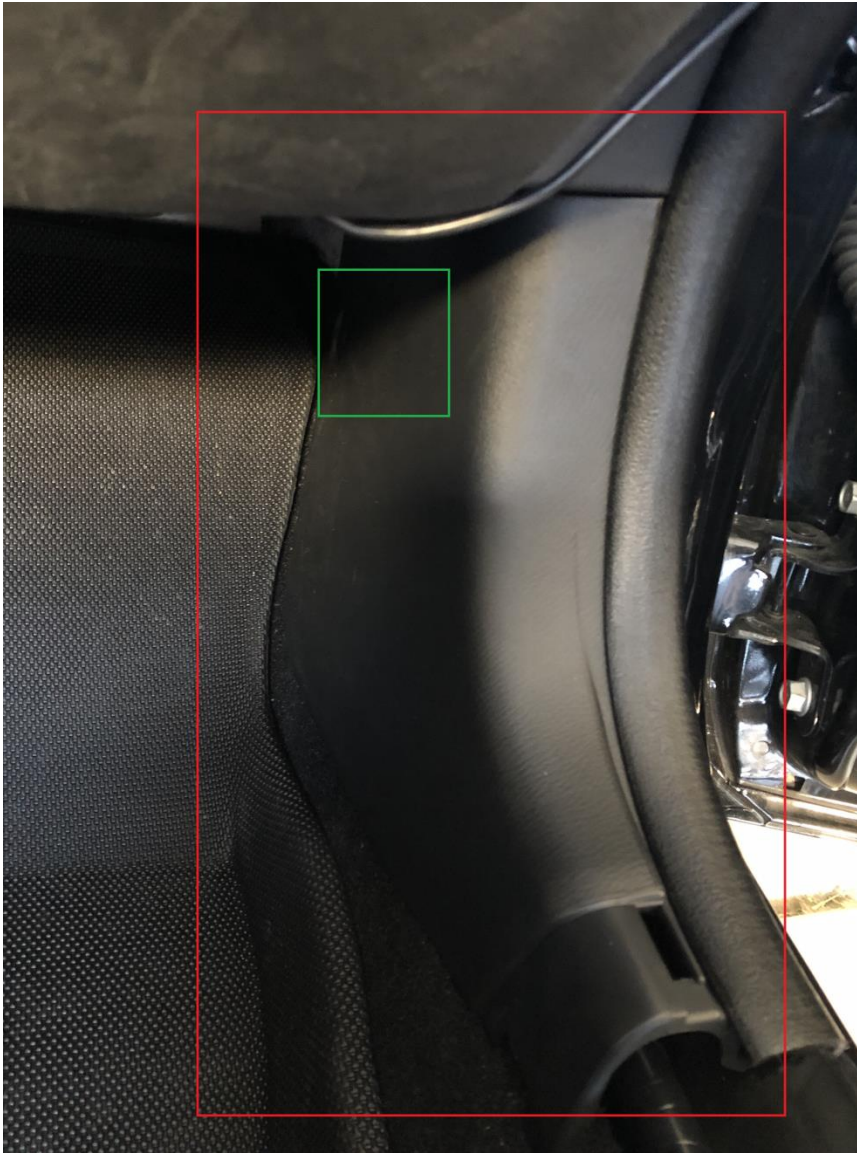
5. Move to the passenger side door and remove the scuff plate. This is just attached with clips. Start at one end, slide your hand underneath, pull up to release the clips at that side and slide your hand forward until all the remaining clips release:



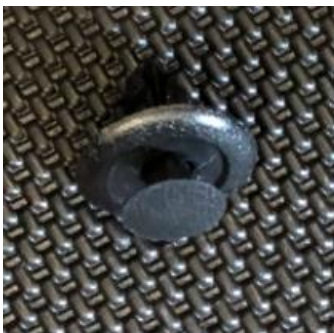
The picture below shows the underside removed piece with the location of the clips:



6. Remove the passenger front side trim. This is the piece in the foot-well shown in the following picture.



The piece is secured by a pop-clip roughly where the green square is. Use a trim tool or flat headed screwdriver to pop up the center section of the clip and then pull the whole clip out.



Put the clip somewhere safe. Note, the center pin is only loosely attached to the outer section – be careful they don't become separated and get lost.

Remove the seaming strip – this just pulls off:



You can now remove the trim piece by pulling it backwards towards the rear of the car. It is secured by a single clip that will just pop out.

7. Remove the shift knob. On the manual cars, it simply unscrews – if you have an automatic, refer to Ameridan's post at <https://21stcenturyfiat124spider.wordpress.com/2017/03/05/removing-the-shift-knob/>



8. Next, remove the center console. This is listed as three separate pieces in the Mazda instructions, but it actually comes out as a single piece – this includes the trim surrounding shifter, the rubber cubby space under the AC controls and the piece containing the command knob controls:



The trim is held in by clips that just pop out. Start by lifting the silver bezel of the shifter surround at the rear passenger side. Once the first clip pops, work your way rearward down the passenger side, around the back of the trim, and up the driver's side until all the clips release (lift the rear cubby lid to get it out of the way).

Once all the clips are released, gently lift the trim and look underneath - there are two cables under the command knob unit that need to be released. Push the retaining clips on the plugs in and slide them out. For some reason, I found it difficult to locate the retaining clip on the larger of the two plugs. The following picture shows where it is:



Once the cables are unplugged, slide the trim up over the shifter and remove from the car.

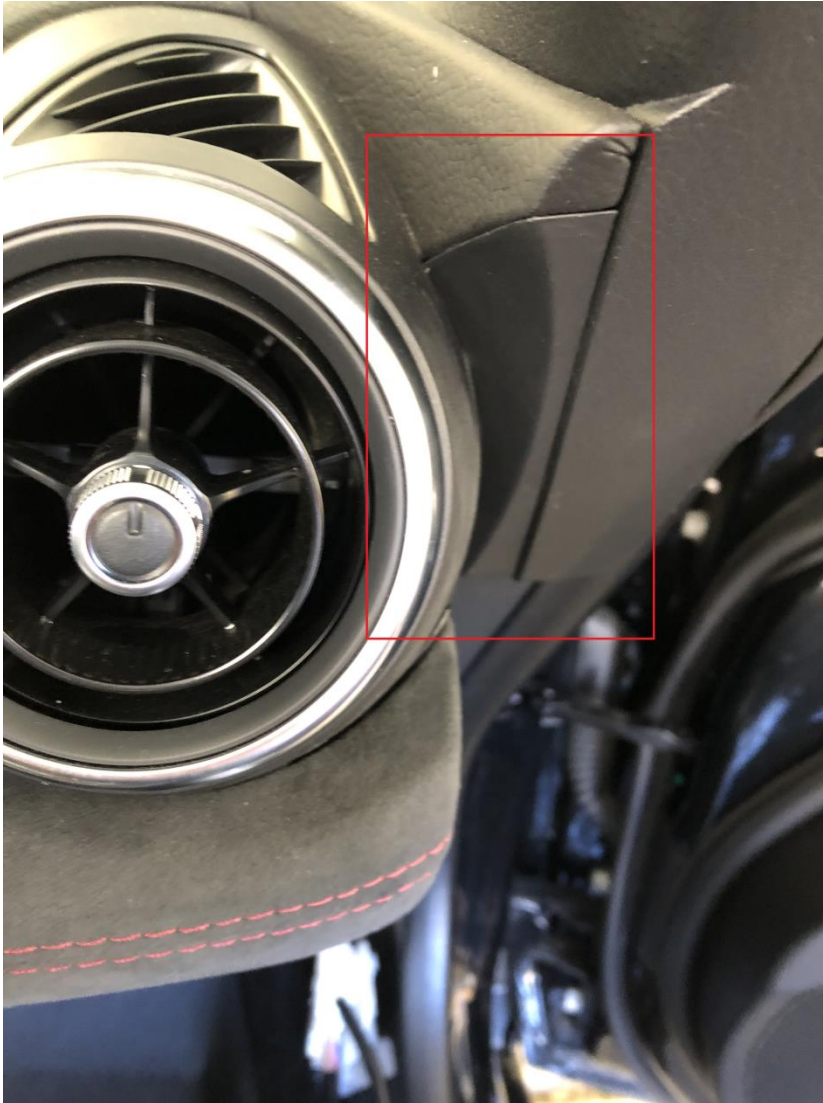
9. Remove the parking brake boot panel. This is held in by two clips – one at the front and one at the rear. I used a trim tool to pop the clips out:



Once both clips are released, slide the boot and trim up and over the brake lever to remove:



10. Remove the passenger side A-Panel trim – this is the panel running up the side of the windscreen. To start, use a trim tool or flat headed screwdriver to remove the piece highlighted below:



This piece should pop out by prying it backwards towards the rear of the car. Once it is removed, remove the rest of the A-Pillar trim by starting at the top and pulling towards the center of the car. Work your way down the trim as each clip pops out in turn. Note that there is a tweeter speaker attached to the trim. Rather than try and release the speaker cable (which is a bit tricky), I just laid the disconnected trim on top of the dashboard. The following picture isn't very clear (lots of reflection from the windscreen), but it's taken from outside the car and shows the trim lying on the dashboard:



11. Next remove the passenger side lower trim:

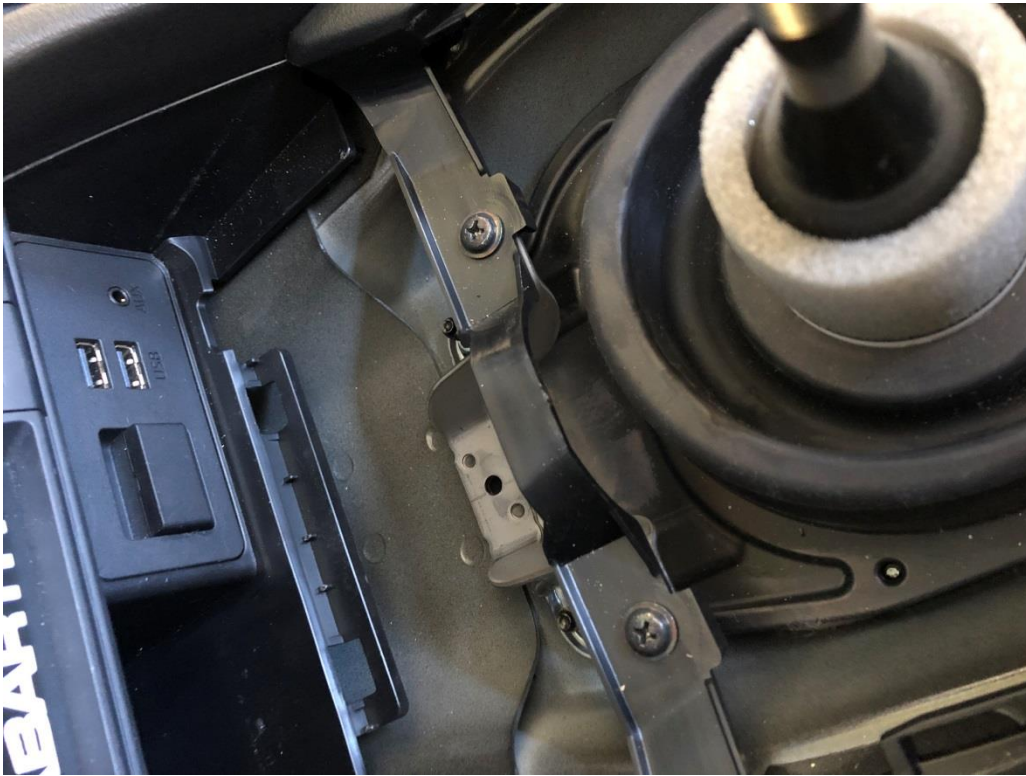


This is held in by a single 10mm bolt and a series of clips that just pop out. The bolt is in the passenger foot-well:



Once the bolt is removed, the trim can be removed by starting at the passenger side and pulling back towards the rear of the car, releasing each clip in turn.

12. Remove the rear center console – this includes the trim surrounding the transmission tunnel and the rear cubby. Start by removing your cupholders if you have them. Next, use a Philips screwdriver to undo the two screws just in front of the shifter:



The panel is now held in by pop out clips. Slide your hands under each side of the front of the panel and lift it upwards to release the clips. Slide your hands backwards down each side of the panel popping each clip out in turn until panel is released:





Remove the panel from the car.

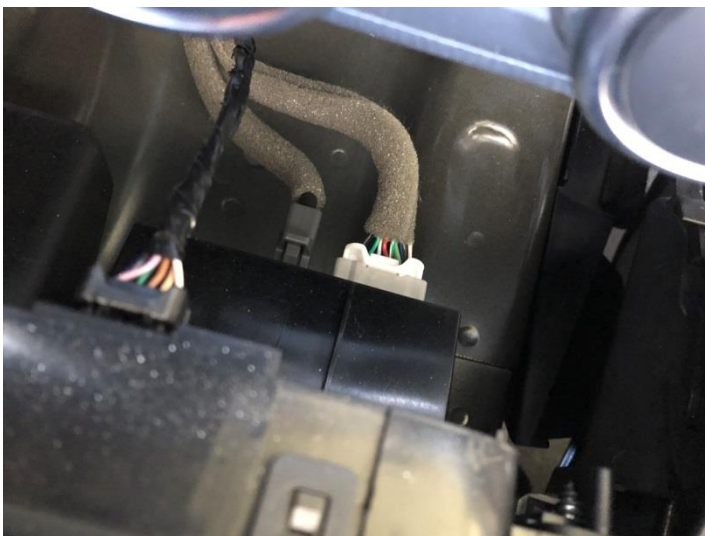
13. Remove the front console and front console panel. The Mazda instructions described this as a two-step process, but I found both pieces came out as a single unit. This is the existing USB hub, the panel with the seatbelt and airbag warning lights, the seat heater buttons and the surrounding trim:



Put one hand on either side of the trim surround and pull back – the entire unit should release as one:



Note that there are three cables on the back of the unit that need to be released. Push down on the locking tab of each and slide it out.



In the picture above, the foam wrapped cable with the small black plug will be replaced with our

two new cables. The cables with the larger black and white plugs will be reused. Take note of which cable is which before moving on. We'll be replacing the USB hub in the removed unit shortly so keep it handy.

14. Remove 'Center Panel No. 2'. This is the panel with the hazard warning lights switch and central air vent in it. It's simply held on with pop out clips. I found it easiest to start at the passenger side of the panel and work my way back to the center of the car popping out each clip in turn:



There is a single cable for the hazard warning lights that needs to be released on this piece.

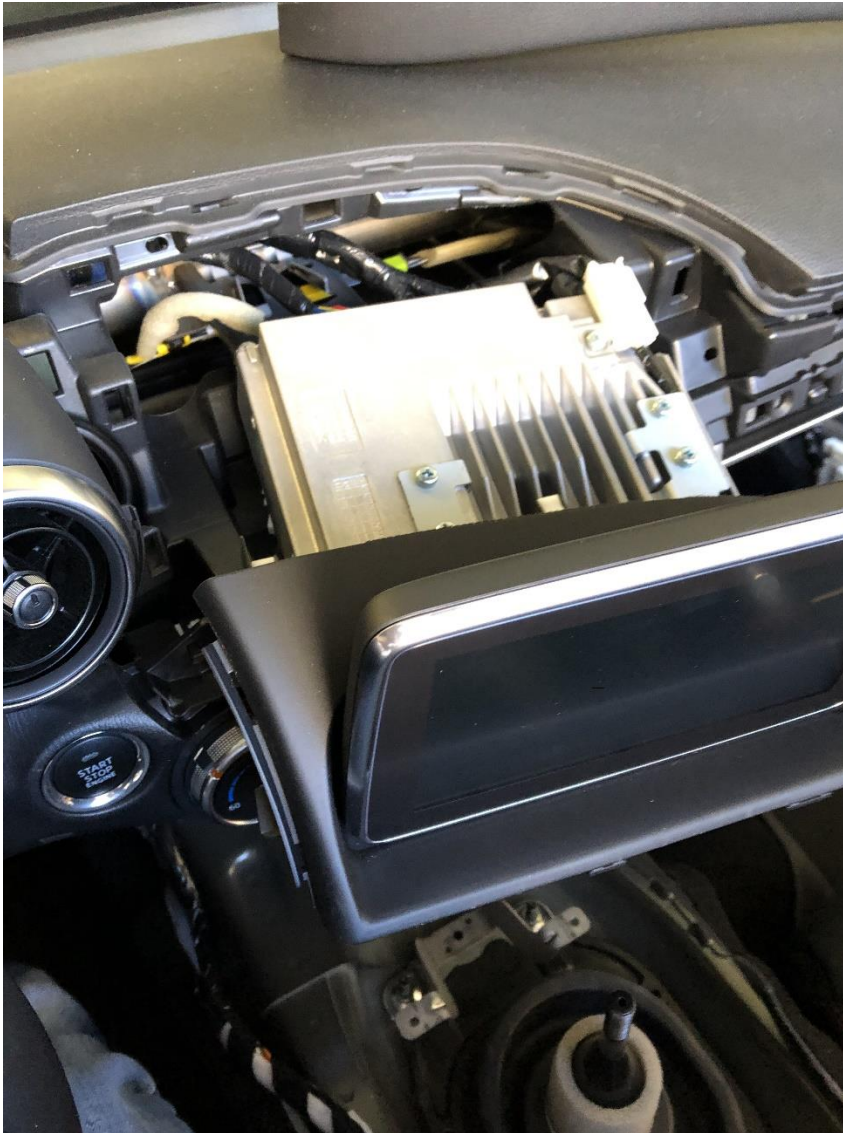
15. Pop out the meter (gauge cluster) hood. Note, you don't need to actually remove this, just pop it out and move it to the side far enough that we can get access to the CMU. Start by lowering the steering wheel as far as possible. You can then pop the hood out by pulling it straight back towards the rear of the car. It is held in by two clips immediately on the steering wheel side of each air vent. Once released, move the hood rearwards and towards the driver side of the car, out of the way of the CMU:



16. Remove the display screen and CMU. This is attached with a single 10mm bolt:



Once the bolt is removed, pull the unit back to release:



There are five cables on the back of the unit that need to be released. The cable with the black and green plug will be replaced by our new cables. The other cables will be reused. Pay attention to which cable comes out of which socket as you are releasing each.

We have now removed all necessary trim pieces.

STEP 2 – USB HUB REPLACEMENT

We'll now replace the original USB hub in the front console with our new unit.

1. Separate the front console trim from the rest of the unit by pulling it backwards until the retaining clips release:



2. Remove the airbag / seatbelt warning light unit by undoing the two Philips screws on each side:

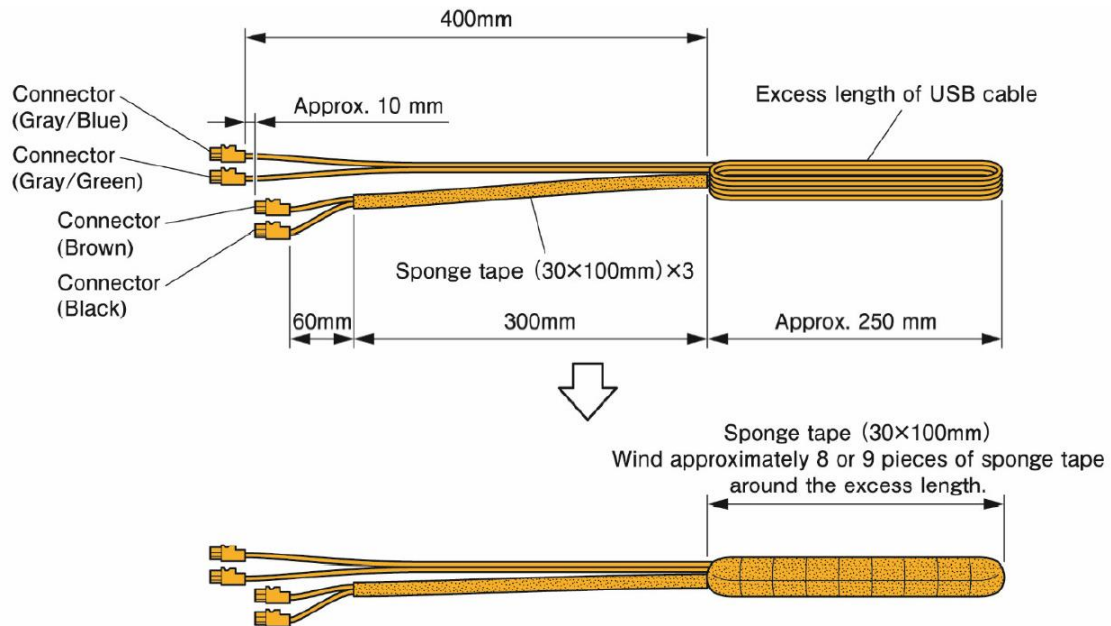




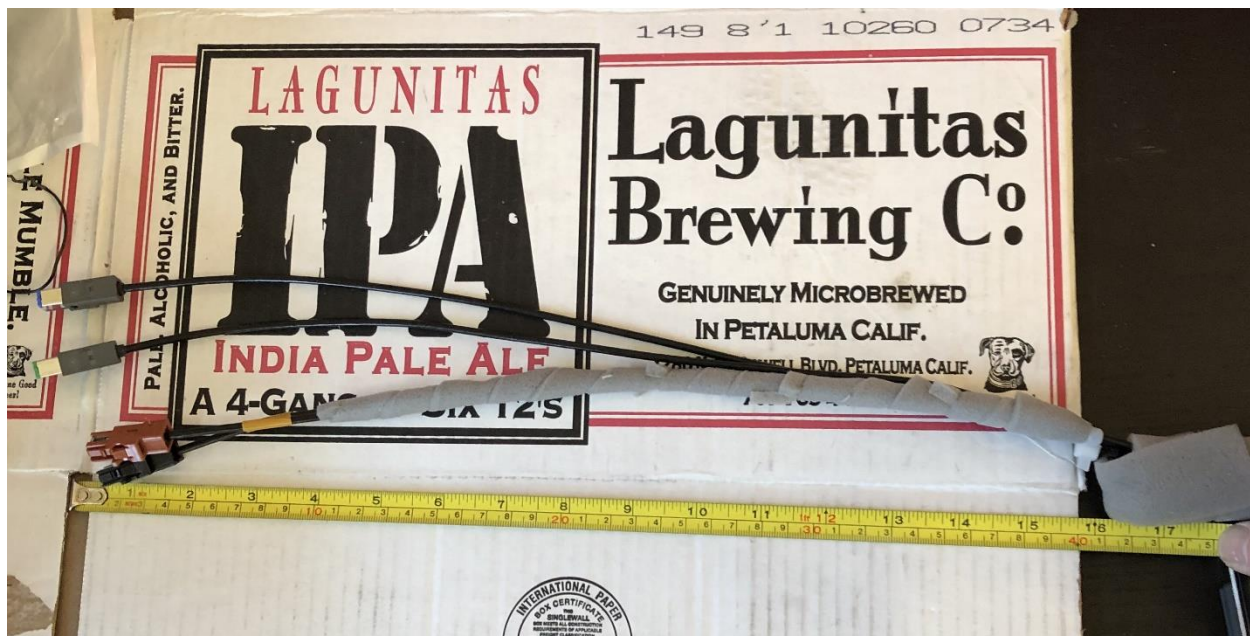
3. Pop out the old USB Hub by pushing down on the 4 retaining tabs, one at a time. Slide the old unit out and then slide the new unit in in its place. (Sorry – I somehow forgot to take pictures of this).
4. Once the new unit clicks in to place, reassemble the front console by screwing the light panel back in place and clicking the trim back on.

STEP 3 – USB CABLE PREPERATION

We're now going to prepare the USB cables for installation by lining up the plugs on each cable, measuring out the required run, and tidying up the excess cable with cable ties and foam. The Mazda instructions provide the following diagram showing what we are trying to do:



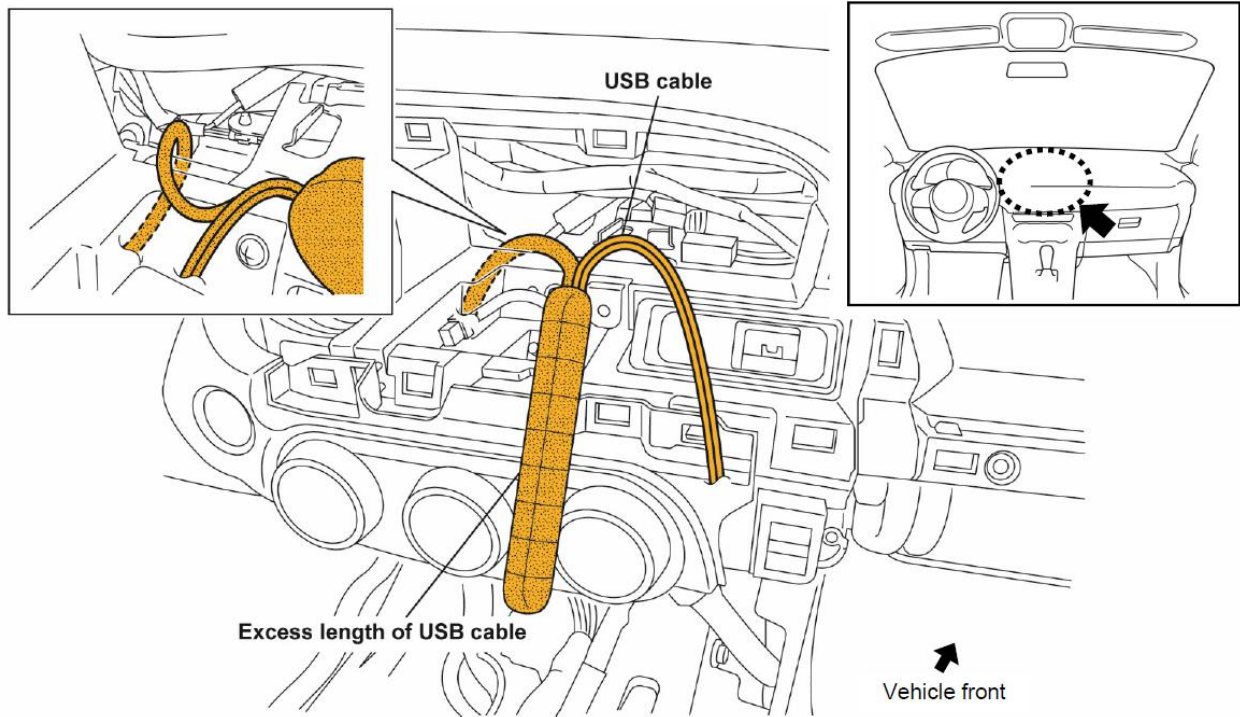
The measurements in this are pretty good. I found by following these I had just enough cable to work with without large amounts of excess in the way as I was reinstalling everything. My prepared cables looked like this:





STEP 4 – USB CABLE ROUTING

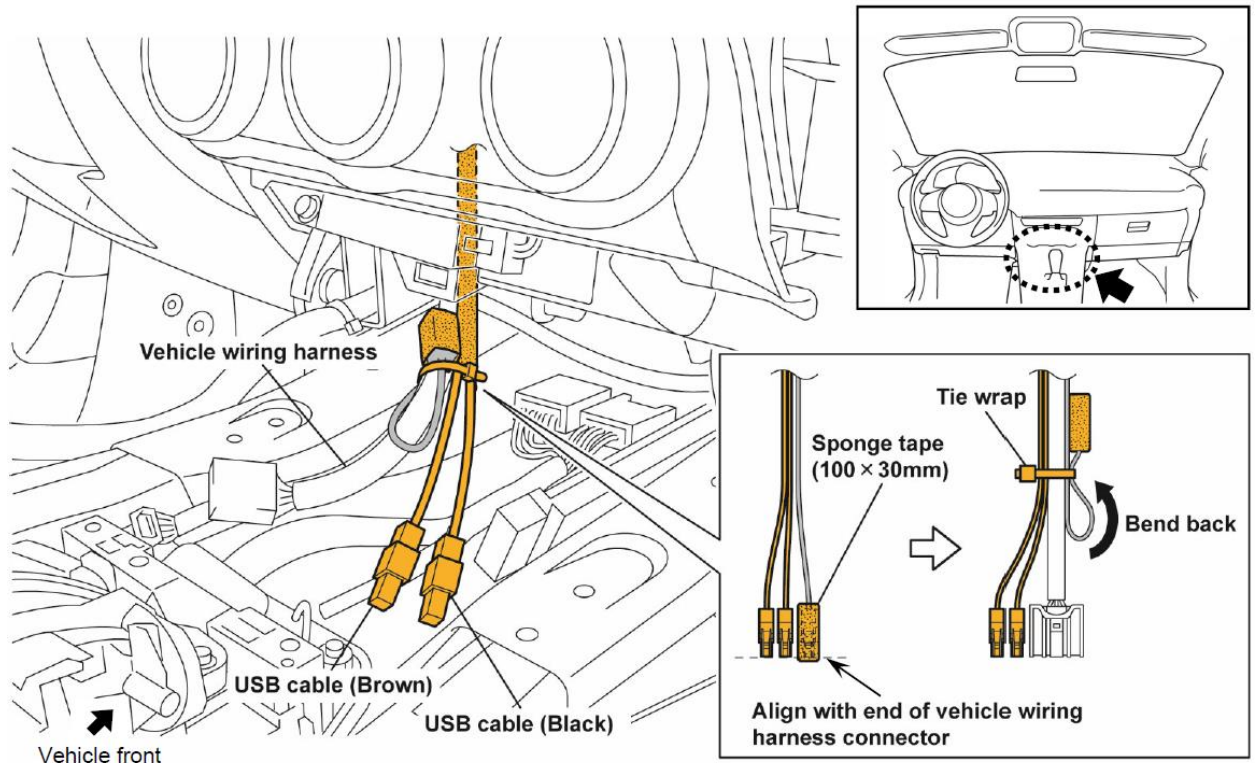
1. Move your prepared USB cables to the car.
2. Route the end of the cables you wrapped in foam tape (with the black and brown connectors) from the top of the dash down to the bottom:



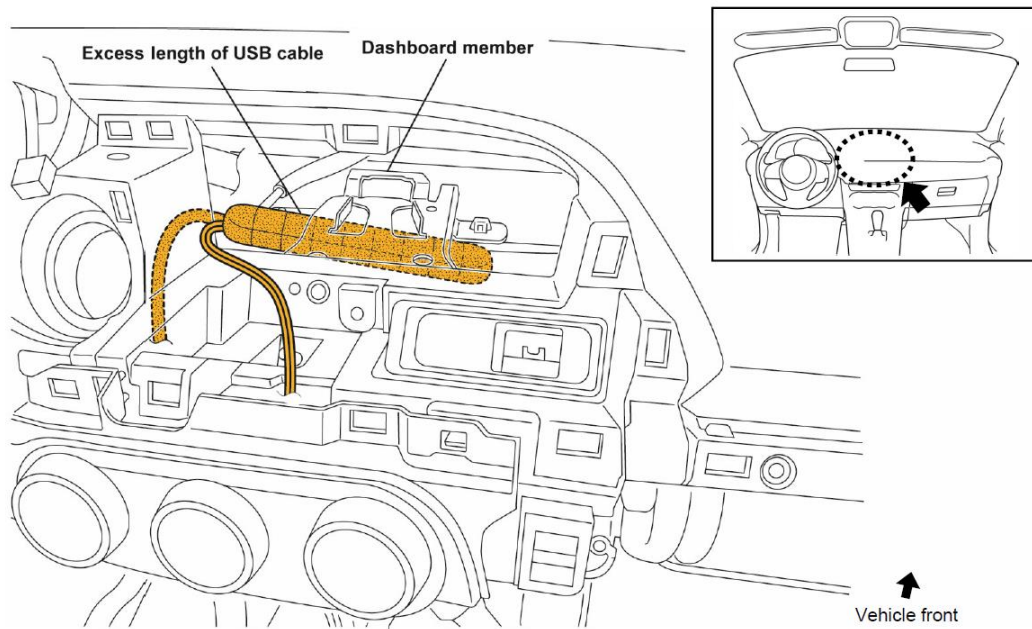
I found there was enough of a gap in the area shown to be able to route the cables down quite easily:



3. Once you have the cables routed, align the new USB cables with the existing cable (black plug with foam wrapped cable), wrap the existing plug with foam so it doesn't rattle against anything, bend it back out of the way, and attach the new cables to the existing wiring harness with a cable tie:



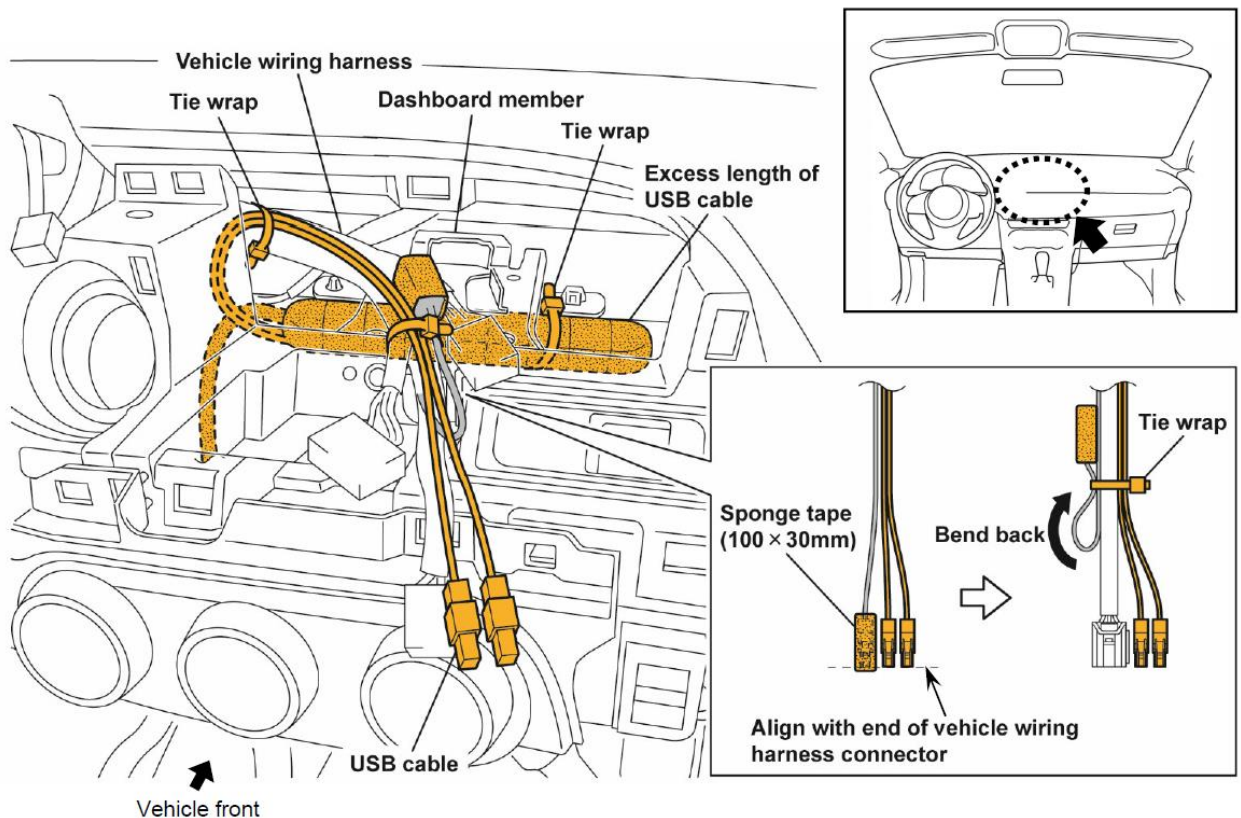
4. Tuck the excess USB cable that you wrapped in foam under the upper dashboard member:

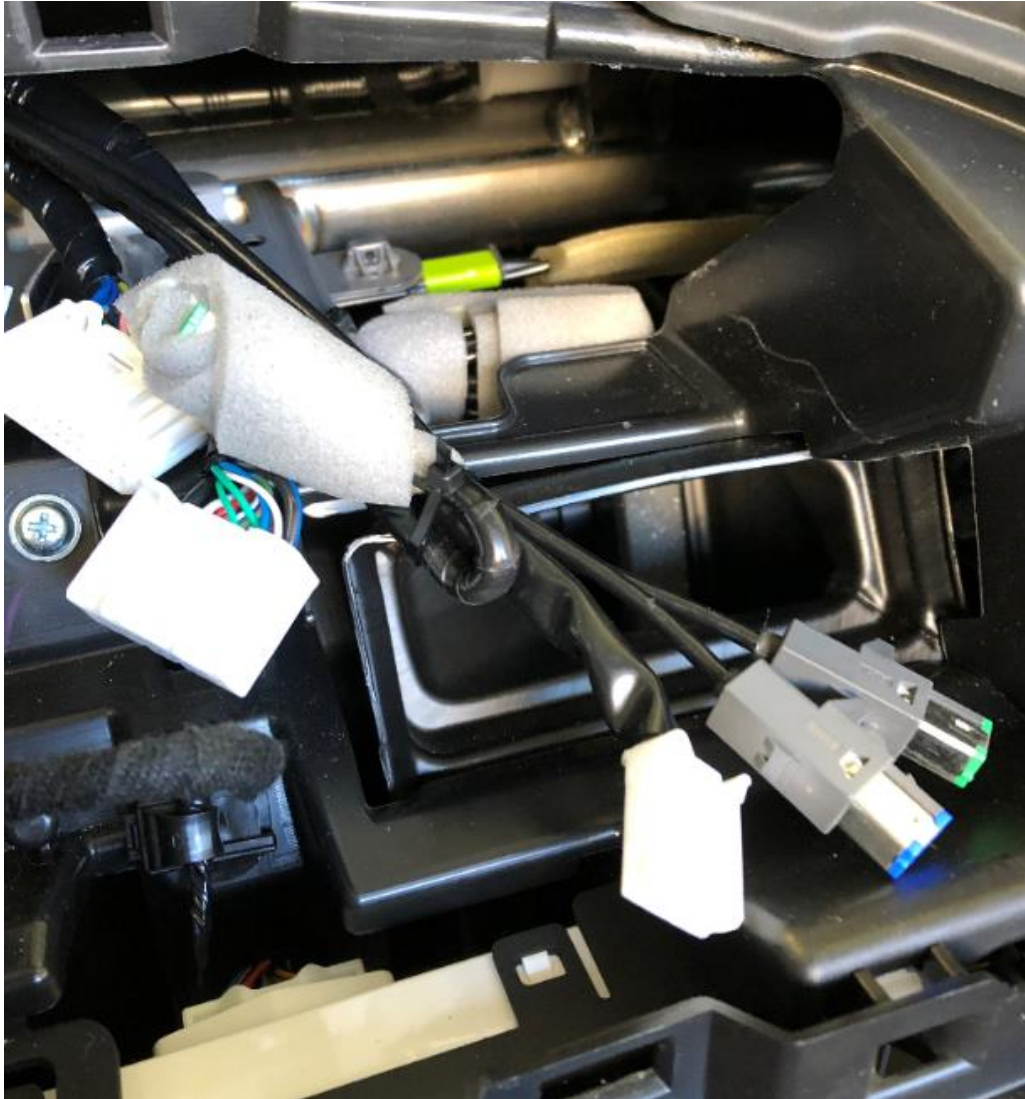


Secure the excess cable to the dash member with a cable tie:



5. As with the lower USB cables, align the new upper USB cables with the existing cable, wrap the existing cable in foam, bend it back out of the way and secure the cables to the existing wiring harness with a cable tie. :

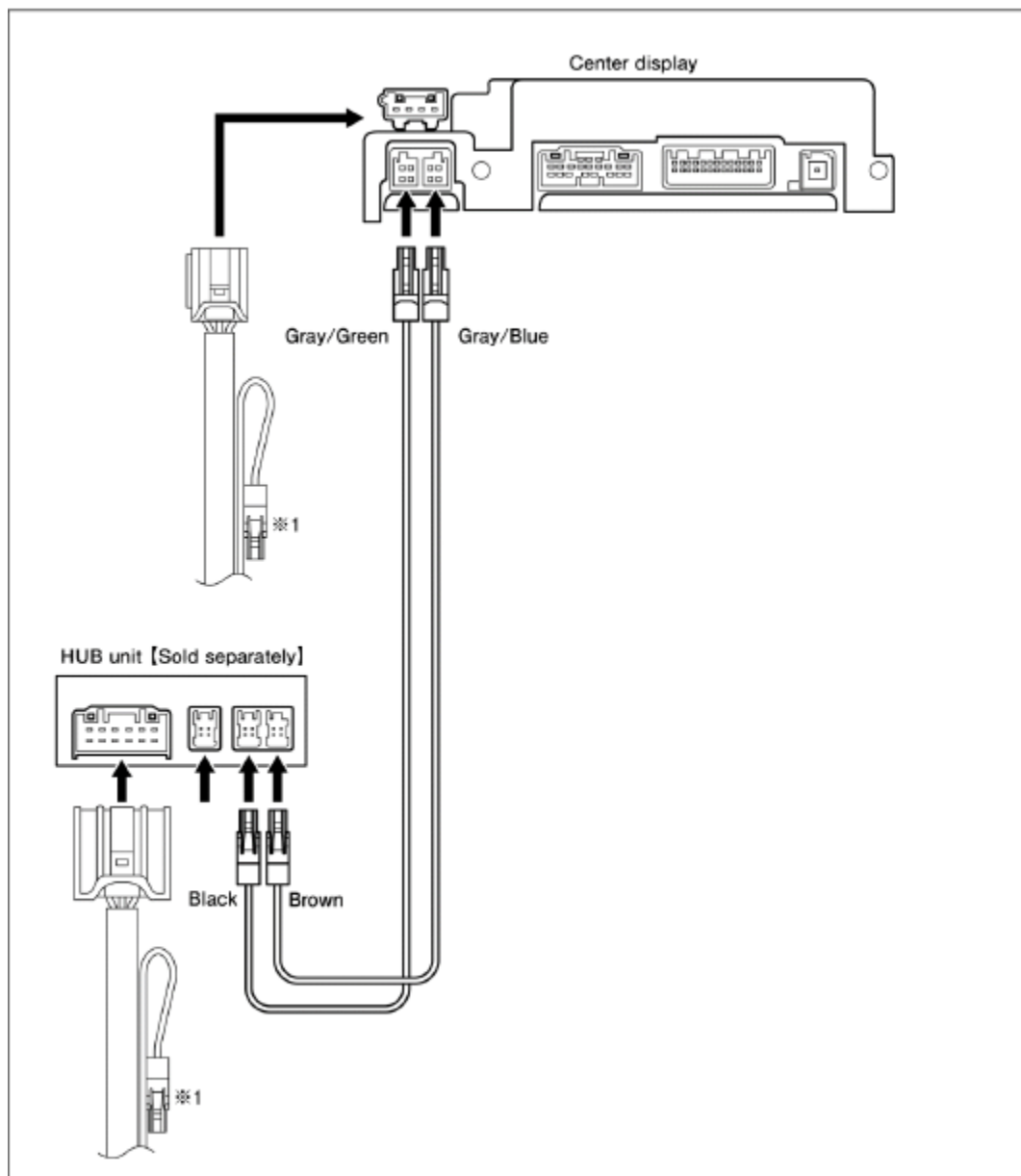




6. Secure any excess length of the new USB cables to the dash member with a cable tie. I didn't grab a photo of this, but you can see it in the wiring diagram above.

STEP 6 – RECONNECT USB HUB AND CMU

1. Place the CMU / display unit and front panel containing the new USB hub close enough to their respective connectors to be able to plug them back in, but do not fully reinstall the units just yet.
2. Connect the USB hub and CMU according to the following diagram. Any cable not on this diagram should be replaced in the plug it was originally removed from. Specifically, there are three additional cables that need to go back in to the CMU and a cable for the airbag /seatbelt warning lights in the front panel. Each plug will only fit in the socket it is intended for.



3. Once everything is plugged in, reconnect the negative terminal on your battery and push the start button once to put the car in accessory mode. Plug your phone into the new USB socket with the smartphone icon and confirm CarPlay or Android Auto starts up (make sure you are running iOS 12 on your iPhone for the best experience. I believe on Android phones you may need to download an app to enable Android Auto, but I'm not familiar enough with it to say for sure):



If everything starts up – congratulations, the install is complete. Switch off the car by hitting start twice then button everything back up by reversing the trim removal instructions. If there are any problems, double check you have plugged the new USB cables in to the correct sockets and try again.